Inborn Errors of Metabolism and Osteopetrosis

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91.1 Inborn Errors of Metabolism

91.1.1 Definition and Epidemiology

Inborn errors of metabolism (IEM) comprise a large group of inherited diseases. They are individually rare. HCT is indicated only in a small number of diseases.

It can be curative in certain lysosomal disorders, peroxisomal disorders or certain disorders of mitochondrial function. This review will be limited to the commoner indications reported in HCT registries and which together account for most of transplanted IEM.

91.1.2 Diagnosis

Timely diagnosis is imperative in IEM since in all such diseases, HCT is better at preventing disease progression than reversing established disease manifestations. Indeed, HCT might be contraindicated in those patients presenting with more advanced disease.

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This is true for all lysosomal storage diseases and for X-ALD.

Diagnosis is made in three ways

- Through early clinical recognition of disease manifestations
- Through screening of pre-symptomatic individuals within a known affected kindred
- Population screening for disease, such as in the neonatal period, with new-born screening

In X-ALD, transplant is indicated for the prevention of progression of cerebral inflammation, which manifests with a high interpersonal variability, even within an affected kindred. Progress of inflammation is detected with serial cerebral MRI scans (at 6 or 12 monthly intervals) in boys, that are genetically affected but currently well.

91.1.3 Classification (See Table 91.1)

91.1.4 Risk Factors

Patient performance score at transplant predicts transplant outcome. Patients with an adverse performance score at transplant also have an inferior long-term survival as the transplant fails in advanced disease to prevent disease progression.

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Table 91.1 Classification of inborn errors of metabolism

IEM	
Hurler syndrome, MPSIH	Hurler syndrome. This is the most severe phenotype of iduronidase deficiency, a lysosomal storage disorder (LSD), which results in the accumulation of glycosaminoglycans, heparan and dermatan sulfate. There is progressive multiorgan dysfunction including psychomotor retardation, severe skeletal disease, life-threatening cardiopulmonary complications, and premature death, HCT prevents early death and attenuates the multisystem disease manifestations as the deficient enzyme is donated by engrafted donor leucocytes to host tissues ("cross-correction")
X-linked adrenoleukodystrophy (ALD)	In this X-linked disorder, there is accumulation of very long-chain fatty acids in the brain and adrenal glands arising from their defective metabolism by a peroxisomal, membrane protein encoded by the ABCD1 gene Clinical manifestations in genetically affected boys are highly variable, even within a kindred. The principle role of HCT is to prevent progression of early cerebral ALD, an inflammatory demyelinating disease of childhood that is seen in about 40% of genetically affected individuals HCT does ***not*** influence other illness such as adrenal insufficiency or the later myeloneuropathy of the spinal cord
Metachromatic leukodystrophy (MLD)	This is an autosomal recessive LSD, and there is accumulation of sulfatides, a myelin component, due to deficiency of the arylsulfatase A enzyme. There is demyelination in the central and peripheral nervous systems, and clinical manifestations are related to residual enzyme activity. In the late infantile disease, the commonest and most severe phenotype, there is progressive neurological dysfunction and early death usually by the age of 4 years HCT is ineffective in preventing progression of early presenting disease, although it may have a greater impact on later, attenuated disease especially when applied early in the course of the illness. HSC GT is recently available for presymptomatic late infantile disease, and corrects disease by delivery of supra-physiological enzyme by leukocytes, derived from gene-modified stem cells

91.1.5 Prognostic Index

Not available.

91.1.6 First-Line Treatment (Summary)

Multimodality therapies are usual in IEM.

- Residual disease manifestations will require management beyond the HCT episode. This will include orthopaedics, ENT, and speech therapy in lysosomal storage disorders (LSDs), as well as family and educational support.
- Pharmacological enzyme replacement therapy (ERT) is used in MPSI but does not correct neurological disease as it does not cross the blood–brain barrier, and alloantibody formation might limit its use in somatic disease. It is used to improve pre-HCT performance, but it has *not* been shown to influence transplant outcomes.

91.1.7 Second-Line Treatment (Summary)

See Sect. 91.1.6., above.

91.1.8 Autologous HCT

Gene-modified auto-HCT approaches have been shown to improve outcomes in late infantile MLD as the graft delivers more enzyme than possible in a conventional HCT. This treatment has been recently licensed by the EMA for therapy in several European countries.

Similar approaches are under investigation for other lysosomal disorders including MPSIIIA, MPSII, Gaucher (particularly where there is neurological involvement, GD3), and MPSIH. It is likely that this approach will greatly change the landscape of therapeutic approaches in metabolic disease in the coming years.

Similar approaches have been successful in X-ALD. The mechanism of action is different, since there is no supraphysiological production of enzyme. Of course, autologous HCT is also safer. Consecutively, no immune suppression is needed with a lower risk of infections and no risk of GVHD.

91.1.9 Allogeneic HCT in MPSIH (Hurler), MLD, and X-ALD (See Table 91.2)

Table 91.2 Main characteristics of allo-HCT for MPSIH (Hurler), MLD, and X-ALD

Indicated in	
MPSIH (Hurler) is a standard indication for HCT In MLD, HCT is usually reserved for later (attenuated) forms of the disease, namely, juvenile and adult forms (note HCT GT is licensed and standard of care in the more severe forms of disease, manifesting earlier) In X-ALD, HCT is indicated in early cerebral inflammatory disease. Ordinarily, a genetically affected individual has serial (6 monthly or annual) MRI scans from early childhood, and HCT is carried out when there are early MRI changes of demyelination (the MRI changes are scored as a Loes score)	
Contraindications	
Where MPSIH is diagnosed late, the opportunity for HCT to meaningfully alter the natural history of the disease might be lost. No predetermined and simple rules can be applied. HCT is usually not offered to a child presenting beyond the age of 30 months, but careful multidisciplinary assessment is required. Not transplanting means the child will die of somatic disease manifestations, and the decision not to transplant is not easy to make Late infantile MLD is not usually considered for allogeneic HCT. Note that such disease—if diagnosed in a timely fashion—has been shown to be markedly improved using an autologous, ex vivo HSC gene therapy approach Advanced cerebral X-ALD is considered a contraindication for HCT. Disease will progress through transplant. The MRI scan-derived Loes score might predict those who will benefit most from HCT	
Donor	
In LSD, non-carrier MFD > MUD > carrier MFD In LSD, UCB is frequently preferred to BM, since the post-HCT chimerism is higher in such recipients, and the interval between referral and HCT is likely shortest (primary rejection might be higher using UCB) PB is rarely used as a donor cell source In X-ALD, MFD > MUD Haplo-HCT is rarely indicated in IEM	
Conditioning: standard	
Engraftment is difficult in IEM. Generally reduced intensity conditioning and ex vivo TCD are associated with high rates of graft loss MSD/MFD: IV BU (MAC AUC)/FLU (160 mg/m^2) MUD: IV BU (MAC AUC)/FLU (160 mg/m^2) AUTO (HCT GT): IV BU (MAC AUC). No immune suppression is necessary	
Conditioning: reduced toxicity	
Occasionally reduced toxicity conditioning might be employed In somatic IEM, such as Wolman or attenuated MPS: TREO (42 g/m^2)/FLU (160 mg/m^2)/TT (10 mg/kg)	
Source of SC	
UCB often preferred in LSD (higher chimerism, and better enzyme delivery therefore) BM rather than PB in MUD donors No ex vivo TCD as this is shown to contribute to graft loss	
GvHD prophylaxis	
MSD/MFD: ATG/Campath, CSA + MMF MUD: ATG/Campath, CSA + MMF Unleated CB: Proximal ATG (−9 to −6), CSA + MMF or CSA + PRD (note rituximab is often added to the preparative regimen in CB transplant to reduce post-transplant autoimmune cytopenia and to reduce primary graft rejection)	
TRM in MPSIH	*OS in MPSIH*
MSD: <5% MUD: <10%	Engrafted survival of >80% and Overall survival of 90%

MAC AUC doses adjusted to achieve MAC AUC, *MSD* match sibling donor, *MFD* match family donor, *MUD* much unrelated donor, *LSD* lysosomal storage disorder, *EIM* inborn errors of metabolism

91.2 Osteopetrosis

91.2.1 Definition and Epidemiology

Osteopetrosis (OP) is a generic name for a variety of rare monogenetic diseases characterized by sclerosis of the skeleton. At least nine variants are known with different modes of inheritance and severity, which cumulatively have an incidence ~1:100,000. The disease originates from reduced or complete lack of osteoclast function and, as a consequence, impairment of bone resorption.

91.2.2 Diagnosis

In addition to the obligate increased bone density of all bones (X-ray), a combination of symptoms can be found in *classical infantile osteopetrosis* after birth. These symptoms include characteristic changes of the head (macrocephalus, frontal bossing, choanal stenosis), vision impairment (due to narrowed foramina), hematological insufficiency (thrombocytopenia, anemia, leukocytosis), hepatosplenomegaly (due to extramedullar hematopoiesis), and hypocalcemia (with secondary hyperparathyroidism). Cave: OP is a genetical and phenotypical heterogenous disease with *atypical presentations* (incomplete and/or delayed onset of symptoms). In these cases, an intensive work-up including spine biopsy and cranial MRI is recommended.

91.2.3 Classification

Osteopetrosis	
Infantile "malignant" autosomal recessive OP (ARO)	Clinical symptoms in infancy, death without HCT usually in the first decade of life, biallelic mutations in *TCIRG1, CLCN7, SNX10, TNFRSF11A/RANK, and FERMT3/KINDLIN-3;* HCT indicated, if excluded: – "Neurodegenerative OP" (all *OSTM1* and about half of *CLCN7* cases) – "Extrinsic osteoclast defects" (*TNFSF11/RANKL cases)*
Intermediate osteopetrosis	Clinical symptoms in the first decade, HCT may be indicated in severe forms with hematological insufficiency and (imminent) visual impairment after individual assessment Specific from: CA2 deficiency (renal tubular acidosis with cerebral calcifications): HCT may be considered after individual assessment in cases with hematological insufficiency and (imminent) visual impairment if neurological and renal problems are controlled
Benign osteopetrosis (ADO)	M. Albers Schoenberg (monoallelic CLCN7 mutations): HCT usually not indicated; HCT may be considered after individual assessment in clinical severe cases

91.2.4 Risk Factors

There is an increased risk of pulmonary hypertension (pre and post HCT) and SOS/VOD (post BMT). The risk of non-engraftment and rejection increases with severity of disease and age.

91.2.5 Prognostic Index

Not available.

91.2.6 First-Line Treatment (Summary)

Symptomatic, steroids may be beneficial to improve hematological symptoms.

91.2.7 Second-Line Treatment (Summary)

Not available.

91.2.8 Autologous HCT

Clinical trials for gene-modified auto-HCT for TCIRG1 defects in preparation.

91.2.9 Allogeneic HCT (See Table 91.3)

Table 91.3 Main characteristics of allo-HCT for osteopetrosis

Indicated in	Infantile osteopetrosis: clinical symptoms and exclusion of neurodegenerative and extrinsic osteoclast defect
Contraindications	Neurodegenerative osteopetrosis: symptoms (non-hypocalcemic convulsions/EEC changes, severe progredient developmental delay) and/or biallelic mutations in *OSTM1* and *CLCN7*; cave: only about half of *CLCN7* mutations cause neurodegeneration Osteopetrosis not intrinsic to defects in differentiation or function in osteoclasts: *TNFSF11/RANKL*
Donor	MFD > MUD > haplo (cord blood not recommended)
Conditioning: standard	MSD/MFD: IV BU (MAC AUC)/FLU (160 mg/m^2) MUD: IV BU (MAC AUC)/FLU (160 mg/m^2) Haplo: IV BU (MAC AUC)/FLU (160 mg/m^2)
Conditioning: reduced toxicity	MSD/MFD: TREO/FLU (150 mg/m^2)/TT (8–10 mg/kg) MUD: TREO/FLU (150 mg/m^2)/TT (8–10 mg/kg)
Conditioning: post cy protocol	In patients with MMUD or haplo donors, an adapted PT-CY protocol should be used (see ESID guidelines. Lankester et al. 2021)
Source of SC	Matched donors, PT-CY protocol: T replete BM > PB
GvHD prophylaxis	MSD/MFD: CSA + MMF (consider ATG or Campath in MFD) MUD: CSA + MMF + ATG (or Campath) Haplo—T replete: Campath, PT-CY, TAC (or CSA) + MMF
TRM	MSD/MUD: 10–20% Haplo: ~20–30%
OS	MSD: ~90% MUD: ~80% Haplo/MMUD: ~70%

MAC AUC doses adjusted to achieve MAC AUC, *MSD* match sibling donor, *MFD* match family donor, *MUD* much unrelated donor, *MMUD* mismatch unrelated donor

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